CAMERA HEALTH PLATFORM

EXECUTIVE OVERVIEW

Automated daily health checks for cameras on moving machinery: what it does, what it saves, and what it needs to go live across all terminals.

MANAGEMENT BRIEFING
VERSION 1.1 · OCTOBER 2026

DP WORLD

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AT A GLANCE

Cameras on cranes, spreaders and trailers are vital for safety and operations, but checking thousands of them by hand every day is slow and inconsistent. The Camera Health Platform takes one picture from every camera each morning, finds the faulty ones automatically and learns from engineers' decisions, so people only spend time on cameras that really need attention.

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camera faults detected automatically

~30 min
to check about 3,600 cameras (default settings)

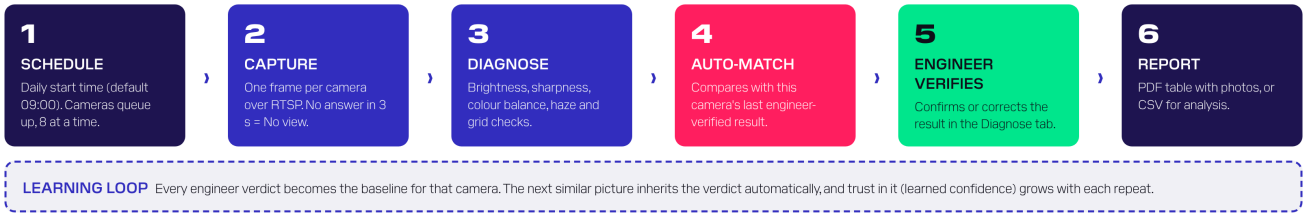
117→4 h
daily person-hours for 1,000 cameras (illustrative)

100%
on-site: no cloud, no internet at runtime

THE PROBLEM

- Cameras are mounted on **moving machinery**, so the view is never the same twice. Standard CCTV health tools that compare against a fixed reference picture do not work.
- Faults such as a dirty lens, a colour tint or a partly covered lens are easy to miss and are often found only when footage is needed after an incident.
- Manually logging in to each camera, saving a screenshot, judging it and writing it up takes several minutes per camera, every day.
- Different camera generations (old 2K units and new 4K units) look different, so one set of rules raises many false alarms.

THE SOLUTION



Daily cycle for each camera

Automatic

Daily schedule, parallel capture, computer-vision diagnosis, confidence score, PDF and CSV reporting. Runs on one PC in the local network.

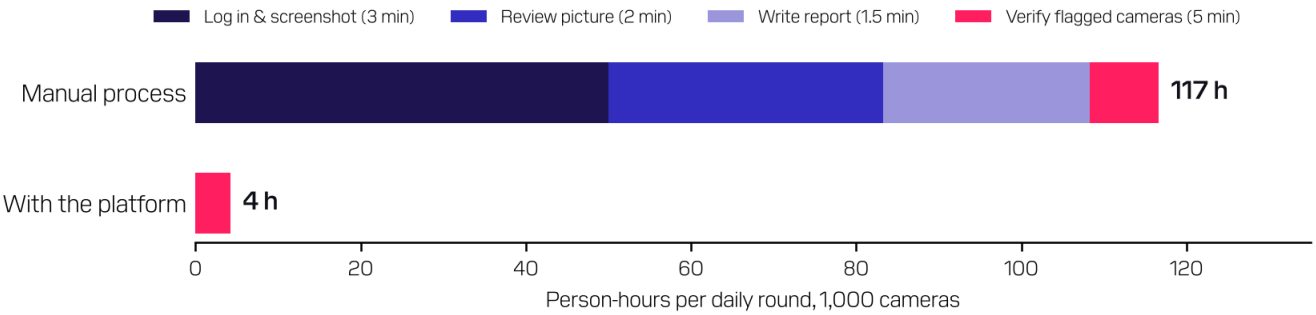
Human in the loop

Engineers confirm or correct each new finding. Every decision is recorded with name, time and remark, giving a full audit trail.

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VALUE

EFFORT COMPARISON



Illustrative, using the platform's built-in role estimates (3 + 2 + 1.5 minutes per camera for capture, review and reporting; 5 minutes to verify a flagged camera), 1,000 cameras and 10% flagged per day, half of them repeats that are matched automatically. Adjust to site figures.

Time

The three clerical steps per camera disappear. Engineers verify only flagged cameras, and repeats of known conditions are handled automatically.

Coverage & consistency

Every active camera is checked every day with the same rules. Nothing depends on who is on shift.

Faster repair

No view, dirty lens and tint problems are found within the morning and reported with a photo, so maintenance can act the same day.

Fewer false alarms over time

Per-camera learning means an old camera that is always slightly soft is accepted once and not flagged again, while real deterioration is still caught ("worse than last check").

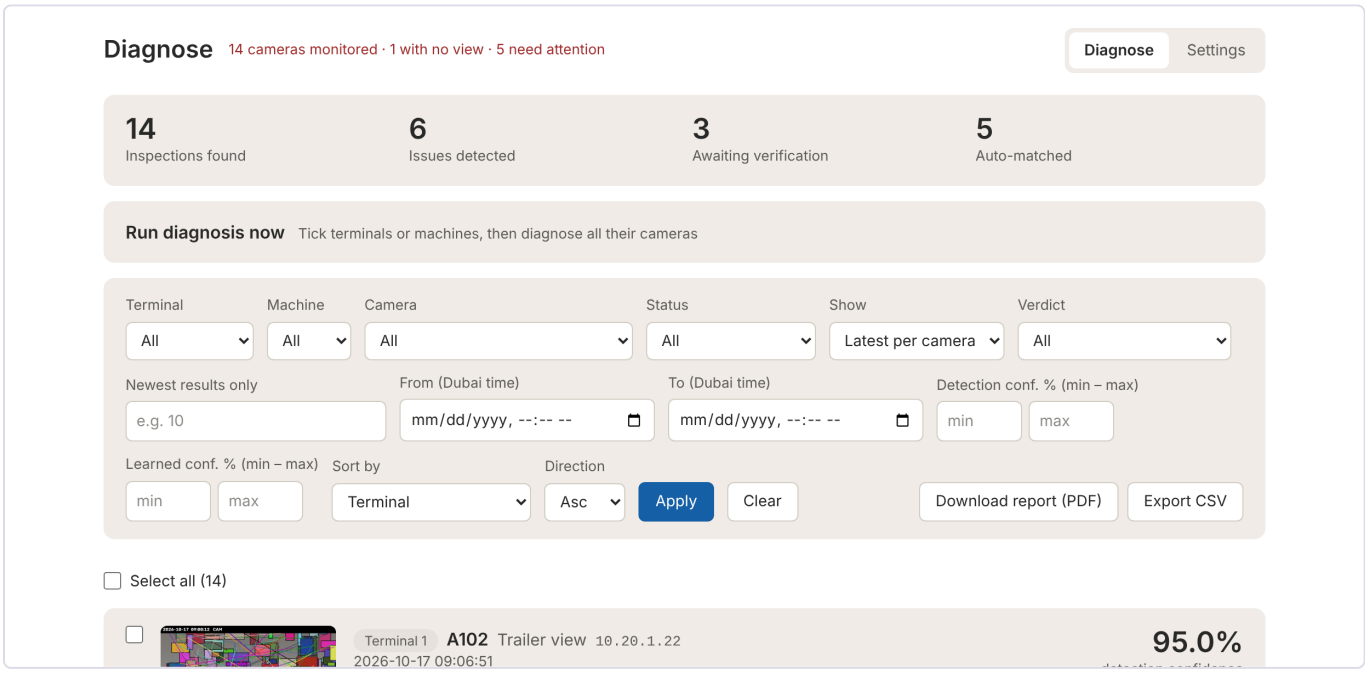
Audit trail

Every inspection keeps its photo, measurements, verdict, engineer name and time. Reports can be produced for any date range, terminal or machine.

Low cost

Open-source components on a single PC. No licences, cloud fees or new hardware at the cameras.

WHAT THE ENGINEERS SEE



The Diagnose tab: summary figures, run-now control, filters and one card per inspection

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CAPABILITIES

CAPABILITY	DETAIL
Fault detection	No view, too dark, overexposed, green / blue / red tint, dirty or blurry lens (including haze), partial occlusion. Each with a 50-99.9% confidence that shows how obvious it is.
Per-camera learning	An engineer's verdict becomes that camera's baseline. Later pictures with the same or a milder condition inherit it automatically; clearly worse ones are flagged for a new look. Trust in an automatic result grows from 68% to 99% with repeats.
Scale	Thousands of cameras. Excel import of 2,500 cameras in about a second; 8 cameras captured in parallel by default (up to 32); two cameras on the same machine never at the same time.
Scheduling	Daily start time, pause between cameras, terminals or machines can be left out; on-demand runs for any terminal or machine with progress and stop.
Configurable checks	Each detection except No view can be switched off centrally (Settings › Schedule › Detections), e.g. while a known site-wide condition is being fixed. Applies to new inspections only.
Reporting	PDF report (one row per inspection with photo, status, verdict and engineer) for any filter or selection, up to 2,000 rows; CSV of all fields for analysis.
History	Removing a camera keeps its history; a replacement with the same names re-links it automatically.
Integration	Read-only status endpoint for the Engineering Building dashboard (status, progress, counts, last error).
Operation	Web dashboard in any browser on the network, Dubai time throughout, dark mode, company logo.

HOW THE LEARNING PROTECTS AGAINST BOTH RISKS

Too many false alarms

Fixed thresholds alone would flag every older camera every day. Learning suppresses known, accepted conditions per camera, so engineers are not flooded and keep trusting the alerts.

Missing a real deterioration

Automatic copies are only made for the same or a milder condition. A lens that gets dirtier, or a new kind of fault, always comes back to a person.

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READINESS & NEXT STEPS

CURRENT STATE

The platform is functionally complete and has been demonstrated end to end with test cameras. Before it is rolled out across all terminals the following items should be closed.

#	ITEM	PRIORITY	WHY
1	Confirm and freeze the production source version	HIGH	The project folder contains files from different versions (database module older than capture module). A clean, tested release is needed.
2	Restrict access and add login	HIGH	The dashboard currently has no sign-in; anyone on the network could change settings or delete data.
3	Protect camera passwords	HIGH	Stored in plain text in the local database.
4	Production web server, auto-start, no-sleep power plan	MEDIUM	The daily run only happens if the platform is running at the start time.
5	Backups and picture retention policy	MEDIUM	About 100-150 MB of pictures per 1,000 cameras per day (roughly 40-55 GB a year) and no automatic clean-up.
6	Agree who may switch detections off, and log it	LOW	Any check except No view can now be switched off for all cameras; changes are not recorded.

PROPOSED ROLLOUT

PHASE	DURATION	SCOPE AND EXIT CRITERIA
1. Pilot	4 weeks	One terminal. Hardening items 1-4 done. Engineers trained. Exit: daily run completes every day; awaiting verification closed daily; false-alarm rate falling week on week.
2. Rollout	4-6 weeks	All terminals imported from Excel, schedule tuned (cameras at the same time) with the network team. Exit: full fleet checked before 10:00 daily.
3. Embed	Ongoing	Weekly KPI review, maintenance tickets linked in remarks, Engineering Building dashboard integration, threshold review each quarter.

KPIS TO TRACK

- **Fleet health:** share of active cameras with final status OK.
- **Backlog:** awaiting verification at the end of each day (target 0).
- **Repair speed:** days from first detection to OK again.
- **Automation rate:** auto-matched share of flagged inspections.
- **False-alarm rate:** share of verdicts that are "Normal / false detection".
- **Run reliability:** daily runs completed on time.

REQUIREMENTS

One always-on Windows PC on the camera network (multi-core CPU, 8-16 GB RAM, SSD sized for picture retention), Python, network access to the cameras' RTSP port (554), a camera account with view rights, and one engineer per shift for verification. Full detail in the Installation & Administration Guide.

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TRAINING & DOCUMENTATION

DOCUMENT	AUDIENCE	USE
Executive Overview (this document)	Management	Decision and rollout planning
User Manual (Word)	Engineers, supervisors	Every screen, the daily routine, verification and reports
Quick Reference Card (PDF)	Engineers on shift	Two printable pages for the desk
Technical & Architecture Guide (Word)	Developers, IT	Design, algorithms, data model, tuning, review findings
Installation & Administration Guide (PDF)	IT, system administrators	Install, configure, secure, back up, upgrade
Training deck (PowerPoint)	Classroom sessions	Management briefing and hands-on engineer training
Training workbook (Excel)	Trainers and trainees	Session plan, competency sign-off, exercises, camera import template

TRAINING PLAN

SESSION	LENGTH	AUDIENCE AND OUTCOME
Management briefing	30 min	Managers: understand value, risks and rollout decisions.
Engineer training	2.5 hours	Verification engineers and supervisors: run the daily routine unaided, give correct verdicts, produce reports.
Administrator training	2 hours	IT: install, schedule, import, back up, upgrade, troubleshoot.
Competency sign-off	On the job, 1 week	Each engineer completes the checklist in the training workbook with a supervisor.